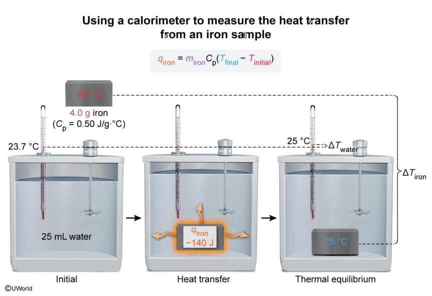
A 4.0 g sample of heated iron was placed into a water bath inside a calorimeter. If 140 J of heat was transferred from the iron to the water and the final temperature of the iron and water was 25 °C, what was the initial temperature of the iron sample before being placed in the water? (Note: The specific heat capacity of the iron is 0.50 J/g·°C.)

- A. 25 °C [2%]
- B. 45 °C [38%]
- C. 70 °C [11%]
- ✔ D. 95 °C [48%]

Correct
48% answered correctly

Explanation:



The specific heat capacity C_p of a substance is the amount of heat required to raise the temperature of 1 g of the substance by 1 °C. Accordingly, if the mass m and change in temperature ΔT of a substance are known, the C_p of the substance can be used to calculate the heat q that is absorbed or released by the substance using the relationship:

$$q = mC_p(T_{\text{final}} - T_{\text{initial}}) = mC_p\,\Delta\,T$$

By convention, a **negative *q*** indicates that **heat is released** from a substance, and a **positive *q*** indicates that **heat is absorbed** by a substance. When two substances are placed in thermal contact, **heat transfers** from the warmer substance to the cooler substance until thermal equilibrium is reached and the two substances have the **same temperature**.

In this question, the iron sample has a negative q because the heat is released from the iron to the water in the calorimeter (ie, q_{iron} = −140 J). After the iron cools in the water, thermal equilibrium is reached and the final temperature of the iron is the same as the final temperature of the water (ie, T_{final} = 25 °C for the iron). Substituting these values, the mass of the iron, and the specific heat capacity of the iron into the heat equation yields:

$$q_{\text{iron}} = m_{\text{iron}}C_p(T_{\text{final}} - T_{\text{initial}})$$

$$-140\text{ J} = (4.0\text{ g})\left(0.50\frac{\text{J}}{\text{g}\cdot{}^\circ\text{C}}\right)(25^\circ\text{C} - T_{\text{initial}})$$

$$-140\text{ J} = \left(2.0\frac{\text{J}}{{}^\circ\text{C}}\right)(25^\circ\text{C} - T_{\text{initial}})$$

Solving for $T_{initial}$ of the iron gives:

$$\frac{-140\,\cancel{\text{J}}}{\left(2.0\frac{\cancel{\text{J}}}{{}^\circ\text{C}}\right)} = (25^\circ\text{C} - T_{\text{initial}})$$

$$-70^\circ\text{C} = 25^\circ\text{C} - T_{\text{initial}}$$

$$T_{\text{initial}} = 25^\circ\text{C} + 70^\circ\text{C} = 95^\circ\text{C}$$

(Choice A) 25 °C is the final temperature of the iron (ie, after cooling in the water); the *initial temperature* of the iron must be higher than its final temperature.

(Choice B) 45 °C is obtained if the negative sign for the released heat is not included in the calculation and ΔT is incorrectly evaluated with T_{final} and $T_{initial}$ reversed (ie, $\Delta T = T_{initial} - T_{final}$).

(Choice C) 70 °C is the change in temperature, not the initial temperature of the iron.

Educational objective:
Multiplying the specific heat capacity by the mass and change in temperature of a sample yields the associated amount of heat absorbed (or released) by the sample. By convention, a negative heat value indicates that heat is released from a substance, and a positive heat value indicates that heat is absorbed by a substance.

Time spent: 0 sec
Subject:
General Chemistry

QID:403208
System:
Thermodynamics, Kinetics & Gas Laws